

Restore an EC2 Instance from the Latest Snapshot

Friday, July 17, 2026 2:45 PM

Task 5: Restore an EC2 Instance from the Latest Snapshot

Objective

Automate disaster-recovery: rebuild an EC2 instance from its most recent EBS snapshot using a Lambda function

Prerequisites

- At least one EBS snapshot exists for the source instance's root volume (pairs well with a "Graded 2" scheduled-snapshot task).
- AWS CLI configured with sufficient permissions to create IAM roles and Lambda functions.

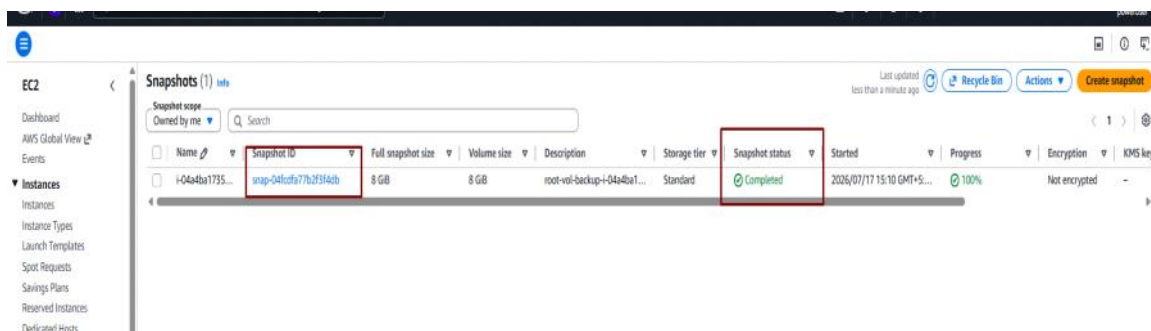
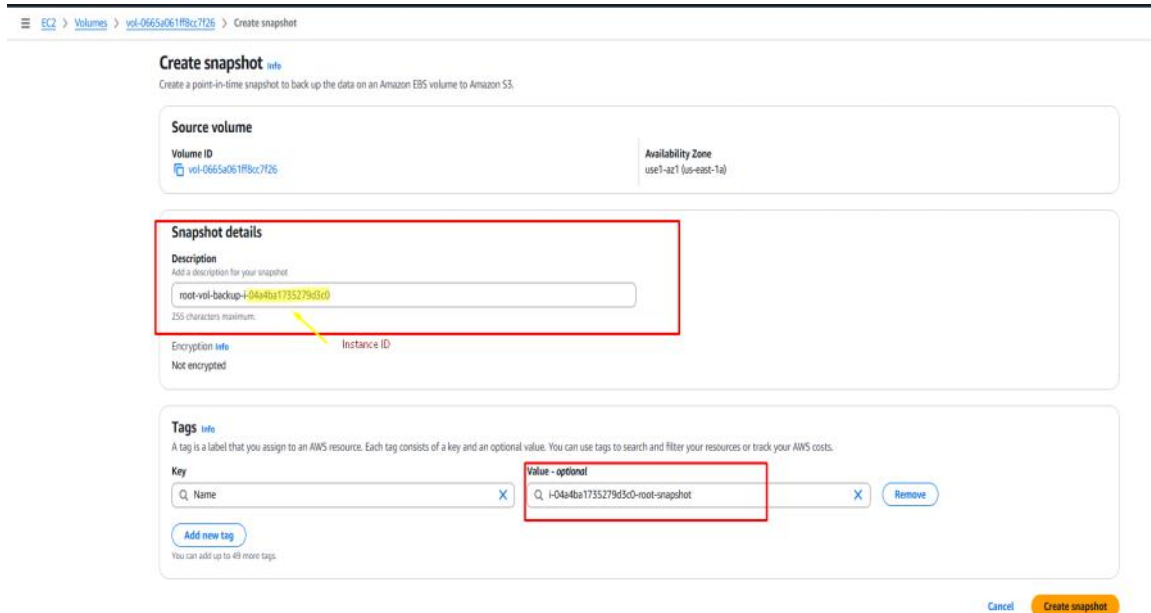
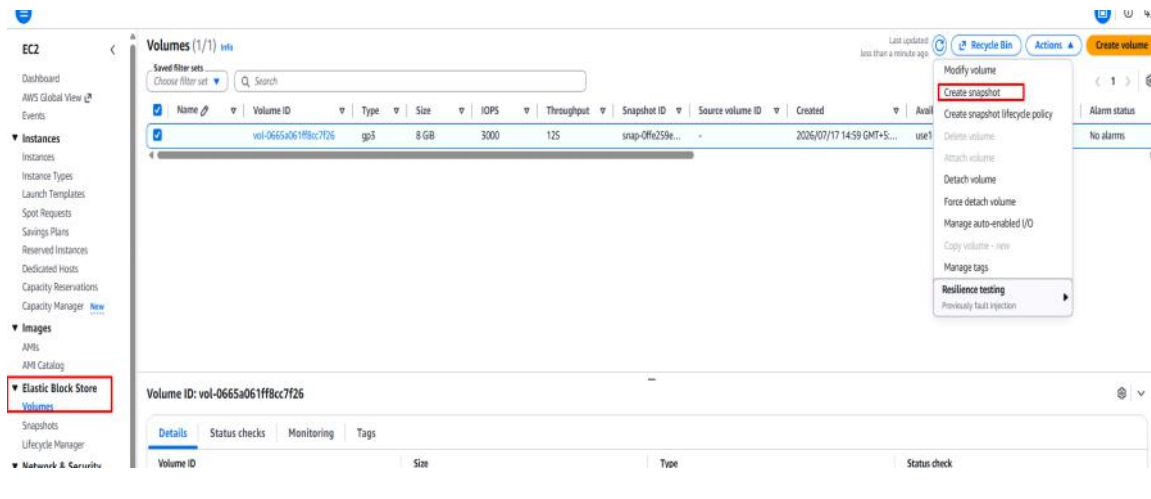
Step 0a — Create a Snapshot of the Source Instance's Root Volume (Console)

- Open the **AWS Console** → search for and open **EC2**.
- In the left sidebar, under **Instances**, click **Instances** → select the source instance (the one you want to protect).
- In the **Storage** tab (bottom panel), note the **Volume ID** **Elastic Block Store**, click **Volumes**.
- Select the checkbox next to that root volume.
- Click **Actions** (top right) → **Create snapshot**.
- In the **Description** field, enter something identifiable, e.g. root-vol-backup-<instance-id>.
- (Optional but recommended) Add a tag: Key = Name, Value = <instance-id>-root-snapshot — makes it easy to filter later.
- Click **Create snapshot**.
- Go to **Elastic Block Store** → **Snapshots** in the left sidebar and confirm the new snapshot shows status **completed** before moving on (large volumes can take a few minutes).

Create EC2 Instance with free tier :

The screenshot displays the AWS Management Console for an EC2 instance. The instance is named 'i-04a4ba7735279d3c0' and is in the 'Running' state. The 'Storage' tab is selected, showing the root volume 'vol-0001a08-0ff8b-7732c' with a size of 8 GiB. The volume is attached to the instance and is in the 'attached' state. The instance is running on an 'm5.xlarge' instance type in the 'us-east-1' region.

Device name	Volume size (GiB)	Volume State	Attachment status	EBS card index	Attachment time	Encrypted	KMS key ID
/dev/sda1	8	in-use	attached	0	2026/07/17 14:55 GMT+5:30	No	-



Step 1 — IAM Role for Lambda

Console steps (detailed navigation)

Create the custom policy:

1. Open the **AWS Console** → search for and open **IAM**.
2. In the left sidebar, click **Policies**.
3. Click **Create policy** (top right).
4. Click the **JSON** tab.
5. Delete the placeholder text and paste the policy JSON shown above.

6. Click **Next**.
7. In **Policy name**, enter EC2RestoreFromSnapshotPolicy.
8. (Optional) Add a description, e.g. "Permissions for Lambda to restore EC2 from snapshot."
9. Click **Create policy**.

Create the IAM role and attach policies:

10. In the left sidebar, click **Roles**.
11. Click **Create role** (top right).
12. Under **Trusted entity type**, select **AWS service**.
13. Under **Use case**, select **Lambda** from the dropdown.
14. Click **Next**.
15. In the policy search box, type EC2RestoreFromSnapshotPolicy → check its box.
16. In the same search box, type AWSLambdaBasicExecutionRole → check its box.
17. Click **Next**.
18. In **Role name**, enter EC2RestoreLambdaRole.
19. Scroll down and review the attached policies (both should be listed) → click **Create role**.
20. Click into the newly created role and copy the **ARN** from the top of the summary page (you'll need it in Step 3) — it looks like arn:aws:iam::<account-id>:role/EC2RestoreLambdaRole.

1. .

The screenshot shows the AWS IAM console 'Create role' wizard, Step 1: Select trusted entity. The 'Trusted entity type' is set to 'AWS service'. The 'Use case' dropdown is set to 'Lambda'. The 'Service or use case' dropdown is set to 'Lambda'. The 'Use case' section shows 'Lambda' selected, with a description: 'Allows Lambda functions to call AWS services on your behalf.' and 'AWSServiceRoleForLambda' listed below it. The 'Next' button is highlighted in yellow.

IAM
Rules
Create role

Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

Maximum 64 characters, use alphanumeric and * - . _ : characters

Description
Add a description for this role.

Maximum 1000 characters, use letters, numbers, and a ., -, numbers 0-9, tabs, new lines, or any of the following characters: * _ : . @ / [] [" ']

Step 1: Select trusted entities

Trust policy

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
8       ],
9       "Principal": {
10        "Service": [
11          "lambda.amazonaws.com"
12        ]
13      }
14    ]
15  }
16 }

```

Step 2: Add permissions

Step 2: Add permissions

Permissions policy summary

Policy name	Type	Attached as
AWSLambdaBasicExecutionRole	AWS managed	Permissions policy

Step 3: Add tags

Add tags - optional [info](#)

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 "Name" tags.

IAM
Roles

Identity and Access Management (IAM)

Dashboard

Access Management

Roles

Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests

Access reports

Access Analyzer
Resource analysis
Unused access
Analyzer settings
Credential report
Organization activity
Service control policies
Resource control policies

Role EC2RestoreLambdaRole created.
View role

Roles (20)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

☐	Role name	Trusted entities	Last activity
☐	AWSServiceRoleForEc2InstanceConnect	AWS Service: ec2-instance-connect	18 days ago
☐	AWSServiceRoleForECS	AWS Service: ecs (Service-Linked Rol	-
☐	AWSServiceRoleForElasticLoadBalancing	AWS Service: elasticloadbalancing	13 days ago
☐	AWSServiceRoleForResourceExplorer	AWS Service: resource-explorer-2	37 minutes ago
☐	AWSServiceRoleForSupport	AWS Service: support (Service-Link	-
☐	AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service	-
☐	demo-function-role-new-lambda	AWS Service: lambda	-
☐	EC2RestoreLambdaRole	AWS Service: lambda	-
☐	Lambda_Demo-role-c20c3amo	AWS Service: lambda	25 days ago
☐	layer-lambda-demo-role-fu5morkq	AWS Service: lambda	25 days ago
☐	new_function-role-bf3fuo9	AWS Service: lambda	22 days ago
☐	new_function-role-jn1uazee5	AWS Service: lambda	22 days ago
☐	new_function-role-wfanyjaut	AWS Service: lambda	24 days ago

Create and attached custom Policy:

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Identity and Access Management (IAM)

Role EC2RestoreLambdaRole created.

EC2RestoreLambdaRole

Allows Lambda functions to call AWS services on your behalf.

Summary

Creation date July 17, 2026, 15:31 (UTC+05:30)	ARN arn:aws:iam:675134942081:role/EC2RestoreLambdaRole
Last activity -	Maximum session duration 1 hour

Permissions | Trust relationships | Tags | Last Accessed | Revoke sessions

Permissions policies (1)

You can attach up to 10 managed policies.

Filter by Type: All types

Policy name	Type	Attached entities
AWSLambdaBasicExecutionRole	AWS managed	2

Permissions boundary (not set)

Generate policy based on CloudTrail events

Related consoles
IAM Identity Center

Step 1: Specify permissions

Specify permissions

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "EC2RestorePermissions",
6       "Effect": "Allow",
7       "Action": [
8         "ec2:DescribeSnapshots",
9         "ec2:DescribeInstances",
10        "ec2:DescribeVolumes",
11        "ec2:RegisterImage",
12        "ec2:CreateImage",
13        "ec2:DescribeImages",
14        "ec2:RunInstances",
15        "ec2:CreateTags"
16      ],
17       "Resource": "*"
18     }
19   ]
20 }
```

JSON | Visual | Actions

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

+ Add new statement

20.2 JSON

9962 of 10240 characters remaining

Security: 0 | Errors: 0 | Warnings: 0 | Suggestions: 0

Step 1
Specify permissions

Step 2
Review and create

Review and create

Review the permissions, specify details, and tags.

Policy details

Policy name

Enter a meaningful name to identify this policy.

EC2RestoreFromSnapshotPolicy

Maximum 128 characters. Use alphanumeric and "+, -, @, _" characters.

Permissions defined in this policy

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it.

Q Search

Allow (1 of 472 services)

Show remaining 471 services

Service	Access level	Resource	Request condition
EC2	Limited: List, Write, Tagging	All resources	None

Cancel Previous **Create policy**

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

Roles

Policies

IAM users

IAM user groups

Identity providers

Account settings

Root access management

Temporary delegation requests

Access reports

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Analyzer settings

Credential report

Organization activity

Service control policies

Resource control policies

Related consoles

IAM Identity Center

AWS Organizations

EC2RestoreLambdaRole

Allows Lambda functions to call AWS services on your behalf.

Delete

Summary

Creation date

July 17, 2026, 15:31 (UTC+05:30)

ARN

arn:aws:iam::675134942081:role/EC2RestoreLambdaRole

Last activity

-

Maximum session duration

1 hour

Permissions Trust relationships Tags Last Accessed Revoke sessions

Permissions policies (2)

You can attach up to 10 managed policies.

Q Search

Filter by Type

All types

Policy name	Type	Attached entities
☐ AWSLambdaBasicExecutionRole	AWS managed	2
☐ EC2RestoreFromSnapshotPolicy	Customer inline	0

Simulate Remove Add permissions

Permissions boundary (not set)

Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

Generate policy

No requests to generate a policy in the past 7 days.

Step 3 — Lambda Function (Boto3)- Package and Deploy

Console steps (detailed navigation)

1. Open the **AWS Console** → search for and open **Lambda**.
2. Click **Create function** (top right).
3. Select **Author from scratch**.
4. **Function name**: RestoreEC2FromSnapshot.
5. **Runtime**: select Python 3.12.
6. **Architecture**: leave as x86_64.
7. Expand **Change default execution role**.
8. Select **Use an existing role**.
9. From the **Existing role** dropdown, select EC2RestoreLambdaRole (created in Step 1).
10. Click **Create function** (bottom right).
11. On the function's page, scroll to the **Code source** panel. Delete the default boilerplate in lambda_function.py and paste in the code from Step 2.
12. Click **Deploy** (above the code editor) to save.
13. Go to the **Configuration** tab → **General configuration** → click **Edit**.

14. Set **Timeout** to 5 min 0 sec (300 seconds — required for the AMI-availability wait).
15. Leave **Memory** at 128 MB (sufficient for this workload).
16. Click **Save**.

≡ [Lambda](#) [Functions](#) Create function

Create function [info](#)

Choose one of the following options to create your function.

☒ **Author from scratch**
Start with a small Python example.

☐ **Use a blueprint**
Build a Lambda application from sample code and configuration artifacts for serverless.

☐ **Container image**
Build a container image to deploy for your function.

Basic information

Function name
Write a name that describes the purpose of your function.

RestoreEC2FromSnapshot 

Function names must be 1 to 64 characters, must be unique to the region, and can't include spaces, non-ASCII characters, or a . At the end, type must be .js and underscore is .

Runtime [info](#)
Choose the language to use to write your function. Note that the supported code runner is only Node.js, Python, and Ruby.

Python 3.12 

Permissions
 By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. To use a different role, choose **Custom execution role** under **Additional settings**.

Custom settings [info](#)

☒ **Durable execution**
Build resilient, stateful applications with automatic failure recovery.

☐ **EC2 capacity provider**
Run Lambda on your preferred EC2 instance types instead of Lambda's serverless compute (default).

▶ **Additional settings**
 Customize your function architecture, execution role, HTTPS endpoints, tenant isolation mode, VPC, code signing, KMS key, and tags.

☐ Cannot add or remove after creation

[Cancel](#) [Create function](#)

Custom settings [info](#)

☒ **Durable execution**
Build resilient, stateful applications with automatic failure recovery.

☐ **EC2 capacity provider**
Run Lambda on your preferred EC2 instance types instead of Lambda's serverless compute (default).

▼ **Additional settings**
 Customize your function architecture, execution role, HTTPS endpoints, tenant isolation mode, VPC, code signing, KMS key, and tags.

General [info](#)

ARM64 architecture
Use ARM64 instead of x86_64 (default).

Custom execution role [Edit](#) 

IAM permissions
[EC2RestoreLambdaRole](#)  attached role to lambda function

Networking [info](#)

Function URL
Assign HTTPS endpoints.

VPC
Connect to a VPC to securely access private resources.

Security & governance [info](#)

Code signing
Ensure only trusted code is deployed.

Encryption with AWS KMS
Use your own KMS key to encrypt deployment package.

Tags

Step 4 — Testing Console

Lambda → RestoreEC2FromSnapshot → **Test** tab → create a new test event:

Add

```
{
  "instance_id": "<Ec2 - instance_id>"
}
```

And ten Test :

Output :

```
{
  "statusCode": 200,
  "new_instance_id": "i-054b8150f186a63c1",
  "ami_id": "ami-08f36c879c6d2dcfc",
  "snapshot_id": "snap-05b6b90cb539294ff"
}
```

Log :

START RequestId: 1d181c07-3b7f-45da-a486-20f494136c0c Version: \$LATEST

Using volume: vol-0178c347a1c255b21, root device: /dev/sda1

Latest snapshot: snap-05b6b90cb539294ff (created 2026-07-17 11:01:33.184000+00:00)

Registered AML: ami-08f36c879c6d2dcfc

New instance launched: i-054b8150f186a63c1

END RequestId: 1d181c07-3b7f-45da-a486-20f494136c0c

REPORT RequestId: 1d181c07-3b7f-45da-a486-20f494136c0c Duration: 2772.23 ms Billed Duration:

3377 ms Memory Size: 128 MB Max Memory Used: 100 MB Init Duration: 604.47 ms

The screenshot displays the AWS CloudWatch console for a function execution. At the top, a green status bar indicates 'Executing function: succeeded'. Below this, the 'Details' section shows the JSON output of the function. The 'Summary' section provides key metrics: Code SHA-256, Function version (\$LATEST), Duration (2772.23 ms), Resources configured (128 MB), Init duration (604.47 ms), Execution time (2 minutes ago), Request ID, Billed duration (3377 ms), and Max memory used (100 MB). The 'Log output' section shows the last 4 KB of the execution log, which matches the output shown in the previous blocks.

```
{
  "statusCode": 200,
  "new_instance_id": "i-054b8150f186a63c1",
  "ami_id": "ami-08f36c879c6d2dcfc",
  "snapshot_id": "snap-05b6b90cb539294ff"
}
```

Summary

Code SHA-256 92jRAQkWRMHHoweahHKtgrOSunpf6Jod41AzZj/MxZzl=	Execution time 2 minutes ago
Function version \$LATEST	Request ID 1d181c07-3b7f-45da-a486-20f494136c0c
Duration 2772.23 ms	Billed duration 3377 ms
Resources configured 128 MB	Max memory used 100 MB
Init duration 604.47 ms	

Log output

The area below shows the last 4 KB of the execution log. [Click here](#) to view the corresponding CloudWatch log group.

```
START RequestId: 1d181c07-3b7f-45da-a486-20f494136c0c Version: $LATEST
Using volume: vol-0178c347a1c255b21, root device: /dev/sda1
Latest snapshot: snap-05b6b90cb539294ff (created 2026-07-17 11:01:33.184000+00:00)
Registered AML: ami-08f36c879c6d2dcfc
New instance launched: i-054b8150f186a63c1
END RequestId: 1d181c07-3b7f-45da-a486-20f494136c0c
REPORT RequestId: 1d181c07-3b7f-45da-a486-20f494136c0c Duration: 2772.23 ms Billed Duration: 3377 ms Memory Size: 128 MB Max Memory Used: 100 MB Init Duration: 604.47 ms
```

Verification

Console:

1. Open **EC2** → left sidebar → **Instances**.
2. In the search/filter bar, filter by tag: RestoredFrom (or just look for the instance named Restored-<snapshot-id>).
3. Confirm **Instance state** shows Running.
4. Select the instance → **Status checks** tab → wait until both checks show 2/2 checks passed.

- To confirm data: select the instance → click **Connect** (top right) → choose **EC2 Instance Connect** or **Session Manager** → **Connect**.
- In the resulting terminal session, verify the expected files/data from the snapshot are present.

EC2 > Instances

Instances (1/2) info

Last updated less than a minute ago

[Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Save filter sets
Choose filter set Find Instance by attribute or tag (case-sensitive)

Instance state = running X [Clear filters](#)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
Restored-snap-05b6b90cb539294ff	i-054b8150f186a63c1	Running	t3.micro	Initializing	us-east-1c	ec2-34-226-190-89.co...	34.226.190.89	-	-
test1	i-001c38c2a2acc873d	Running	t3.micro	3/3 checks passed	us-east-1b	ec2-3-85-165-249.com...	3.85.165.249	-	-

CloudWatch > Log management > [aws/lambda/RestoreEC2FromSnapshot](#) > 2026/07/17/[LATEST]:c7060146523f47a8d7eb4cfc8942548

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

Clear 1m 30m 1h 12h Custom UTC

Timestamp	Message
2026-07-17T11:04:03.159Z	INIT_START Runtime version: python:3.12.mainline.v18 Runtime version ARN: arn:aws:lambda:us-east-1:1:runtime:f2f1ee454341c37ac7910af0b0172f63d62828615853c8c6c7476440f93c66
2026-07-17T11:04:03.767Z	START RequestID: 1d181c87-3b7f-45de-a486-28f494136c8c Version: LATEST
2026-07-17T11:04:04.162Z	Using volume: vol-0178c347a1c255023, root device: /dev/sda1
2026-07-17T11:04:04.239Z	Latest snapshot: snap-05b6b90cb539294ff (created 2026-07-17 11:01:33.184000+00:00)
Latest snapshot: snap-05b6b90cb539294ff (created 2026-07-17 11:01:33.184000+00:00)	
2026-07-17T11:04:04.735Z	Registered API: api-00f9ac879cd2dcfc
Registered API: api-00f9ac879cd2dcfc	
2026-07-17T11:04:06.505Z	New Instance launched: i-054b8150f186a63c1
New Instance launched: i-054b8150f186a63c1	
2026-07-17T11:04:06.548Z	END RequestID: 1d181c87-3b7f-45de-a486-28f494136c8c
END RequestID: 1d181c87-3b7f-45de-a486-28f494136c8c	
2026-07-17T11:04:06.548Z	REPORT RequestID: 1d181c87-3b7f-45de-a486-28f494136c8c Duration: 2772.23 ms Billed Duration: 3377 ms Memory Size: 128 MB Max Memory Used: 100 MB Init Duration: 604.47 ms
	No newer events at this moment. Auto retry paused. Resume